

FM v2.8

001: # Peripheral Neuroimmune and Nociceptive Gene Signatures in Fibromyalgia: A Targeted Reanalysis of Public Transcriptomic Cohorts Informed by UK Biobank Plasma Proteomics 002: 003: — 004: 005: **Authors:** Cristóbal Muñoz Rojas¹ 006: 007: ¹ Independent Researcher, Santiago, Chile. Correspondence: cristoe4@gmail.com 008: 009: **Preprint — Draft v2.8 — August 2026** 010: 011: — 012: 013: ## Abstract 014: 015: Fibromyalgia (FM) is a prevalent chronic pain condition whose molecular basis remains poorly defined. A genome-wide association study (GWAS) meta-analysis of 2.5 million individuals identified 26 risk loci enriched in brain tissues, prioritizing neural and synaptic genes including *DRD2*, *MDGA2*, *CAMKV*, and *CELF4* (Kerrebijn et al., 2025; PMID 41001472). Separately, the largest population-scale plasma proteomics studies of chronic pain (UK Biobank Olink) found no support for the classical IL-6/IL-8 inflammatory axis and instead triangulated CA14 (carbonic anhydrase XIV) as a causal protein in chronic widespread pain via Mendelian randomization and colocalization (Chen et al., 2025, *Adv Sci*; PMID 41025730). Here we perform a targeted, hypothesis-driven reanalysis of two public transcriptomic datasets — GSE221921 (PBMCs, 96 FM / 93 HC) and GSE67311 (whole blood, 67 FM / 75 HC) — interrogating GWAS neural genes, the opioid/tachykinin neuropeptide axis, UKB causal genes, and mast cell markers. In GSE221921 PBMCs we find: (1) *MDGA2* ($q = 1.1 \times 10^{-7}$) and *DRD2* ($q = 2.9 \times 10^{-5}$) robustly upregulated across all five sensitivity models; (2) a **complete opioid/tachykinin neuropeptide circuit activated** — receptor *TACR1* (NK1, Substance P receptor; FC = 2.73, $d = +0.60$, the largest effect size in the panel), *OPRM1* (μ -opioid receptor; FC = 2.28, $d = +0.53$), ligand *TAC1* (Substance P; FC = 2.10, $d = +0.47$) and *OPRK1* (κ -opioid; FC = 1.78, $d = +0.38$), with coordinated co-expression ($\rho = 0.31$ - 0.63) — the largest effect-size block, surviving all five sex-sensitivity models but **not** adjustment for cell composition (0 of 7 axis genes survive; §3.2), which identifies this axis as a **shift in circulating leukocyte populations rather than per-cell transcriptional upregulation**; (3) **two extracellular-matrix / neurite-outgrowth genes from the chronic-widespread-pain causal set** — *COL9A1* (FC = 2.32, $d = +0.88$) and *PTN* (FC = 2.91, $d = +0.61$) — survive Bonferroni correction on the sex-adjusted model and are the only signals in this investigation that also survive cell-composition adjustment (COL9A1 $p =$

0.012-0.020, PTN $p = 0.028$ -0.046 across four deconvolution implementations), making this co-expressed module ($r = 0.51$) — not the neuropeptide axis — **the most robust finding of the investigation**; it is directly relevant to joint pain and nerve repair and detectable in plasma by Olink; and (4) **CA14 — the top UKB causal protein for chronic widespread pain — upregulated in FM PBMC mRNA in the unadjusted analysis** ($FC = 2.29$, $p = 0.0003$, $d = +0.41$) **but not surviving the sex-stratified primary model** (female-only $p = 0.135$), while being **downregulated in plasma** in the UKB cross-sectional analysis, with MR indicating a protective effect of genetically elevated CA14 — relegated here to a reference candidate rather than the primary validation target. Together these signatures reframe the peripheral biology of FM as **extracellular-matrix / neurite-outgrowth dysregulation (COL9A1-PTN) as the cell-intrinsic core, accompanied by a compositional shift in neuropeptide-expressing leukocyte populations, immunomodulation (UKB immune genes) and pH/nociception dysregulation (CA14)** — and we recommend prioritizing **COL9A1 and PTN** (Olink, plasma-detectable, Bonferroni-surviving, composition-robust) in the confirmatory plasma validation, with CA14 measured as a direction-specific reference. Fold changes throughout are ratios of arithmetic group means on FPKM; because the distributions are heavy-tailed, geometric-mean equivalents are substantially smaller (e.g. TACR1 $2.73 \rightarrow 1.21$; COL9A1 $2.32 \rightarrow 1.72$) and both are reported in Tables 2 and 4. Standardized effect sizes (Cohen's d) are unchanged or larger under $\log_2$ transformation. In GSE67311 (whole blood), these PBMC-derived proxies do not replicate in absolute expression (TAC1/OPRM1/IL6 $FC \approx 1.0$); deconvolution rules out neutrophil-driven dilution; the co-expression structure of the axis is partially preserved (5 of 10 pairs show FM > HC, but 2 reverse with HC > FM) — the circuit is transcriptionally coherent in FM, while its amplitude is compartment-dependent. These findings argue that the peripheral molecular signature of FM is dominated by **neuropeptide nociceptive signaling and extracellular-matrix dysregulation**, not classical inflammation, and support prioritizing COL9A1 and PTN for validation in plasma (Olink) with cell-type-resolved, medication-stratified cohorts. 016: 017: **Keywords:** fibromyalgia, COL9A1, PTN, CA14, TACR1, OPRM1, tachykinin, opioid, extracellular matrix, UK Biobank, Olink, PBMCs, targeted reanalysis, nociception, immunomodulation 018: 019: — 020: 021: ## 1. Introduction 022: 023: Fibromyalgia (FM) affects 2-4% of the global population, manifesting as chronic widespread pain, fatigue, cognitive dysfunction, and sleep disturbances (Sarzi-Puttini et al., 2020). Despite its prevalence, FM lacks specific diagnostic biomarkers and its pharmacotherapy remains limited to three FDA-approved medications — pregabalin, duloxetine, and milnacipran — none designed to target FM-specific molecular pathology (Chinn et al., 2016). 024: 025: A recent GWAS meta-analysis by Kerrebijn and colleagues, comprising 54,629 cases and 2,509,126 controls across 11

cohorts, identified 26 genome-wide significant risk loci for FM, with heritability enriched exclusively in brain tissues and neuronal cell types (Kerrebijen et al., 2025; PMID 41001472; medRxiv doi: 10.1101/2025.09.18.25335914). The prioritized genes span dopaminergic signaling (*DRD2*), synaptic plasticity (*CAMKV*, *CELF4*), neural cell adhesion (*NCAM1*, *MDGA2*), axon guidance (*DCC*), and other neural functions (*GPR52*, *HTT*). While the GWAS authors describe these as a neural/CNS network, we note that only *DRD2* is strictly dopaminergic; the others are more broadly neural or synaptic. 026: 027: This genetic architecture motivates a direct question: **are these GWAS-prioritized genes differentially expressed in FM patients?** Several bioinformatic studies have analyzed the GSE221921 PBMC dataset (Mohapatra et al., 2024; Bi et al., 2024; Zhao et al., 2025; Gowri Gopal et al., 2026), but all employed unbiased genome-wide approaches (DEG → PPI → hub genes). None tested the specific hypothesis that the GWAS-defined gene set is coordinately altered — a targeted, hypothesis-driven analysis that is distinct from and complementary to unbiased discovery. 028: 029: In parallel, population-scale plasma proteomics has reframed the peripheral biology of chronic pain. Li ZY et al. (2025; PMID 40048323) profiled 2,923 plasma proteins (Olink Explore) in 51,644 UK Biobank participants with chronic pain, identifying 474 pain-associated proteins and 10 proteins validated as causal by Mendelian randomization — none of them classical inflammatory cytokines. A dedicated FM/nociplastic analysis of the UKB Olink data (Chen et al., 2025, *Adv Sci*; PMID 41025730; 29,254 participants, 2,920 proteins) built multi-protein scores with AUC 0.856–0.880 for CWP, identified 18 proteins with causal relevance by MR, and triangulated **CA14 (carbonic anhydrase XIV)** as the top causal protein by MR + colocalization (PP.H4 > 0.5). Notably, CA14 is **downregulated in plasma** in the cross-sectional analysis while genetically elevated CA14 is protective by MR; and the authors propose **CA14 agonists, not the inhibitor sulthiame**, as the more promising therapeutic direction. Critically, IL-6, IL-8/CXCL8, TAC1 and Substance P appear **zero times** in the largest published chronic pain proteomics screen — the classical inflammatory axis is not supported at population scale. 030: 031: In this study, we: (1) test the GWAS neural gene set for differential expression in PBMCs (GSE221921) and whole blood (GSE67311), including comprehensive sensitivity analyses for sex confounding; (2) extend the analysis to the **opioid/tachykinin neuropeptide axis** (ligands *TAC1*, *PENK*, *PNOC*, *POMC*; receptors *TACR1*, *OPRM1*, *OPRK1*, *OPRD1*); (3) test the **UKB causal genes** (CA14, TNFRSF1B, CD74, COL18A1, BTN2A1, TNFRSF4, CD302, TNFRSF9) in our PBMC data to connect population-scale causal proteomics with transcriptomics; (4) include a negative control panel of mast cell markers; and (5) contextualize our findings with a targeted literature review of dopamine agonist trials in FM and the pharmacology of CA14. 032: 033: — 034: 035: ## 2. Methods 036: 037: ### 2.1 Gene Set Definition 038: 039: **GWAS-Prioritized Neural Gene Set.** We extracted 13 genes from the

26 GWAS risk loci reported by Kerrebijn et al. (2025) that are functionally linked to neural or synaptic signaling: *DRD2*, *NCAM1*, *GPR52*, *CAMKV*, *CELF4*, *DCC*, *MDGA2*, *NPY*, *KYNU*, *SRD5A2*, *PPP2R2B*, *NPC1*, and *HTT*. We designate this set as “GWAS neural genes” rather than “dopaminergic network,” as only *DRD2* is strictly dopaminergic. 040: 041: **Opioid/Tachykinin**

Neuropeptide Axis. Ligands: *TAC1* (Substance P), *PENK* (enkephalins), *PNO*C (nociceptin), *POMC* (β -endorphin).

Receptors: *TACR1* (NK1), *OPRM1* (μ -opioid), *OPRK1* (κ -opioid), *OPRD1* (δ -opioid). HGNC symbols verified (*TAC1* = Substance P; *PENK* = proenkephalin — a distinct opioid ligand, not Substance P; correction documented in

AUDITORIA_INTEGRIDAD_PROXY.md). 042: 043: **UKB Causal**

Gene Set. Genes encoding proteins with MR-supported causal relevance for chronic pain / chronic widespread pain in UKB Olink proteomics, measured in GSE221921: *CA14* (top-ranking causal for CWP, MR + colocalization; Chen et al., 2025) and the immune-signaling genes *TNFRSF1B*, *CD74*, *COL18A1*, *BTN2A1*, *TNFRSF4*, *CD302*, *TNFRSF9* (MR-validated in the multisite chronic pain analysis; Li ZY et al., 2025). *LEP* and *TNF* are not measurable in GSE221921. Note: in the published Chen et al. (2025) analysis, 18 proteins have MR causal relevance for CWP (*CA14*, *COL9A1*, *CRELD1*, *DPEP1*, *LEG1*, *LGALS3*, *MLN*, *PRSS53*, *TNF*, *BPIFB2*, *CTSO*, *DDR1*, *FAM171B*, *IFI30*, *LRRC37A2*, *PTN*, *SFTPD*, *ST3GAL1*); *CA14* is the top-ranking one and is **downregulated in plasma** in the cross-sectional analysis, with MR indicating a protective effect of genetically elevated *CA14*. 044: 045: **Mast**

Cell / Basophil Panel (negative control). Four genes (*CPA3*, *MS4A2*, *FCER1A*, *HDC*) previously identified as differentially expressed in FM whole blood (Jones et al., 2016). These are primarily expressed by basophils and mast cells, which are depleted during PBMC isolation. 046: 047: *NPY* was included in the pre-specified gene set but was absent from the GSE221921 expression matrix. FDR correction was therefore applied to 16 measured genes (12 GWAS neural + 4 mast cell) in the primary GWAS-neural analysis; the expanded panel of 19 genes used for the opioid/tachykinin and UKB-causal analyses was corrected with Bonferroni $\times 19$ or $\times 9$ as specified in §2.3. 048: 049: ### 2.2

Transcriptomic Datasets 050: 051: **GSE221921 (PBMCs, RNA-seq).** FPKM-normalized expression values from peripheral blood mononuclear cells of 96 FM patients and 93 healthy controls

(Mohapatra et al., 2024; PMID 38366049). Sample metadata and expression matrices were obtained from the GEO supplementary file GSE221921_FM_ProcessedData.xlsx. 052: 053: **Critical note on**

sex distribution: The GSE221921 cohort has a severe sex imbalance — FM group: 91 female / 5 male; HC group: 41 female / 52 male. This confounds any unadjusted FM vs. HC comparison, as a portion of the observed signal may reflect sex differences rather than disease effects. We address this through multiple sensitivity analyses (§2.4). 054: 055: **GSE67311 (Whole blood,**

microarray). Affymetrix Human Gene 1.1 ST array expression data from whole blood (PAXgene tubes) of FM patients and

healthy controls (Jones et al., 2016; PMID 27157394). **Metadata verification (2026-08-03): the GEO sample metadata contains 67 FM / 75 HC (142 total), which differs from the 70/70 reported in the original paper.** Pre-computed differential expression results ($\log_2FC$, p-value, FDR-adjusted p-value) were used, with the verified group counts. 056: 057: ### 2.3 Statistical Analysis 058: 059: All p-values from the 16 measured genes in the GWAS-neural analysis were jointly corrected using the Benjamini-Hochberg (BH) procedure (Benjamini & Hochberg, 1995) at $\alpha = 0.05$. Genes were considered significant at $q < 0.05$. 060: 061: For the expanded panel of 19 genes (validate_fm_biomarkers_iter2.py, v3 post-adversarial-audit), expression values are heavy-tailed and non-normal (Shapiro-Wilk rejects normality for all genes, both groups), so **Mann-Whitney U tests** were used as the primary inference, with **Bonferroni correction across the 19-gene panel** ($N_GENES_PANEL = 19$), and **Cohen's d** (pooled SD) reported as the effect size. T-tests are retained only as reference. Classification: proxy significant if $p_Bonf < 0.05$; effect size graded small ($d < 0.3$), small-medium ($0.3 \leq d < 0.5$), medium ($0.5 \leq d < 0.8$). 062: 063: For the UKB-causal gene analysis (9 measurable genes in GSE221921), Bonferroni correction $\times 9$ was applied to the Mann-Whitney p-values. 064: 065: For the opioid/tachykinin axis, Mann-Whitney U p-values are reported; the four core genes (TACR1, OPRM1, TAC1, OPRK1) survive conservative Bonferroni correction across the 21-gene union panel (19 + TACR1 + OPRK1; e.g., TACR1 $p = 0.0010 \times 21 = 0.021$; OPRK1 $p = 0.0017 \times 21 = 0.036$). Spearman rank correlation was used for co-expression analysis across all 189 samples. 066: 067: ### 2.4 Sensitivity Analyses for GSE221921 068: 069: Given the sex imbalance, we applied five analytical models to each gene: 070: 071: 1. **Welch t-test on raw FPKM** (original analysis) 072: 2. **Welch t-test on $\log_2(FPKM+1)$** (variance-stabilizing transform) 073: 3. **Mann-Whitney U test** (non-parametric, distribution-free) 074: 4. **OLS regression: $\log_2(FPKM+1) \sim case + sex$** (sex as covariate, full cohort) 075: 5. **Female-only subgroup: Welch t-test on $\log_2(FPKM+1)$** (91 FM vs. 41 HC, eliminates sex confound entirely) 076: 077: Each model produced p-values that were independently FDR-corrected across all 16 genes. Genes were classified by robustness: "Robust" ($q < 0.05$ in all 5 models), "Supported" ($q < 0.05$ in 3-4 models), "Model-sensitive" ($q < 0.05$ in 1-2 models), or "Not significant." 078: 079: **Definition of the robustness denominator (clarified 2026-08-04).** An intermediate analysis file (analysis/E7_sensitivity_master_GSE221921.csv) scores robustness over a *different* five-model set in which a **male-only** subgroup replaces model 2. That set is not used here: the male-only stratum contains **5 FM patients**, so it has no power to reject any null and cannot function as a robustness criterion — every gene in the panel fails it, including genes that are unambiguously significant under all other models. Throughout this manuscript "Robust (n/5)" refers **exclusively** to models 1-5 above. Where earlier versions reported "Robust 4/5" for COL9A1/PTN, that

score was computed against the E7 set; under the definition used here both genes are **Robust 5/5** (§3.4.5). 080: 081: **Model 6 — cell-composition adjustment (added 2026-08-04)**. Because bulk PBMC expression confounds per-cell transcription with cell-population shifts, we added a sixth model applied to the neuropeptide axis and the COL9A1-PTN module: **OLS** $\log_2(\text{FPKM}+1) \sim \text{case} + \text{sex} + \text{cell fractions}$, where fractions are marker-based estimates for 12 leukocyte populations (§2.2 markers; scripts/deconvolution_cell_types.py). Because fractions are compositional (sum to 1), the largest-mean population is dropped as reference. Verdicts were confirmed across four implementations (NNLS-estimated vs. marker-score fractions $\times$ dropping the reference vs. retaining all 12). To calibrate how much signal this adjustment removes for reasons unrelated to biology, the same model was applied to 600 randomly selected expressed genes (cell-type markers excluded) as a negative control. Script: scripts/e1_deconvolution_adjusted_model.py; results: analysis/E1b_DECONVOLUTION_COL9A1_PTIN.md. 082: 083: ### 2.5 Targeted Literature Review 084: 085: We searched PubMed (terms: “pramipexole fibromyalgia,” “ropinirole fibromyalgia,” “dopamine agonist fibromyalgia”), ClinicalTrials.gov, and the GSK Study Register for clinical trials of dopamine D2/D3 agonists in FM; and PubMed/EuropePMC for CA14, sulthiame, and UKB Olink chronic pain proteomics (Li ZY et al., 2025; Chen et al., 2025). Search date: August 2026. This is a targeted narrative review, not a formal systematic review; no PRISMA protocol was registered. 086: 087: ### 2.6 Limitations of the Analytical Approach 088: 089: We acknowledge that the use of FPKM values with parametric tests is a simplified approach. The gold standard for RNA-seq differential expression is count-based modeling (DESeq2/edgeR/limma-voom; Love et al., 2014). Only FPKM values were available in the GEO supplementary materials; raw counts were not accessible. All analyses were performed in Python 3.10 using pandas 1.5, scipy 1.10, and statsmodels 0.13. The primary inference therefore uses non-parametric Mann-Whitney U with Bonferroni correction (§2.3), which is robust to the FPKM distribution. 090: 091: — 092: 093: ## 3. Results 094: 095: ### 3.1 MDGA2 and DRD2 Are Robustly Upregulated in FM PBMCs 096: 097: Table 1 presents the sensitivity analysis for all 16 measured genes. Two genes — *MDGA2* and *DRD2* — were significant ($q < 0.05$) across all five analytical models and are classified as **Robust**: 098: 099: **Table 1. Sensitivity analysis: q-values (FDR-corrected) across five analytical models.** 100: 101: | Gene | Category | FM mean | HC mean | Welch FPKM | Welch $\log_2$ | Mann-Whitney | OLS sex-adj | Female-only | Robustness | 102: | — | — | — | — | — | — | — | — | — | — | 103: | *MDGA2* | GWAS Neural | 2.582 | 1.069 | 1.1×10^{-7} | 1.0×10^{-8} | 1.1×10^{-7} | 6.3×10^{-6} | 5.5×10^{-6} | **Robust (5/5)** | 104: | *DRD2* | GWAS Neural | 0.721 | 0.271 | 2.9×10^{-5} | 2.6×10^{-6} | 1.6×10^{-6} | 4.7×10^{-4} | 3.6×10^{-5} | **Robust (5/5)** | 105: | *CAMKV* | GWAS Neural | 0.656 | 0.288 | 3.2×10^{-3} | 1.6×10^{-3} | 8.3×10^{-3} | 0.112 | **0.032** |

Supported (4/5) | 106: | *CELF4* | GWAS Neural | 1.341 | 0.848 | **0.046** | **0.013** | **0.010** | 0.167 | **0.032** | Supported (4/5) | 107: | *HTT* | GWAS Neural | 20.60 | 25.17 | 0.120 | **7.5×10^{-3}** | **4.6×10^{-3}** | 0.177 | **0.042** | Supported (3/5) | 108: | *NCAM1* | GWAS Neural | 4.375 | 3.234 | 0.087 | **0.021** | **0.010** | 0.186 | 0.114 | Model-sensitive (2/5) | 109: | *DCC* | GWAS Neural | 2.747 | 1.905 | 0.087 | **0.023** | **0.016** | 0.167 | 0.055 | Model-sensitive (2/5) | 110: | *SRD5A2* | GWAS Neural | 0.187 | 0.139 | 0.251 | 0.145 | 0.092 | 0.167 | **0.042** | Model-sensitive (1/5) | 111: | *GPR52* | GWAS Neural | 2.723 | 3.792 | 0.099 | 0.367 | 0.590 | 0.955 | 0.817 | NS | 112: | *NPC1* | GWAS Neural | 13.01 | 14.49 | 0.546 | 0.108 | 0.083 | 0.335 | 0.268 | NS | 113: | *KYNU* | GWAS Neural | 3.401 | 3.987 | 0.490 | 0.180 | 0.139 | 0.564 | 0.544 | NS | 114: | *PPP2R2B* | GWAS Neural | 2.445 | 2.587 | 0.883 | 0.825 | 0.581 | 0.955 | 0.817 | NS | 115: | *HDC* | Mast Cell | 5.274 | 7.991 | 0.666 | 0.103 | 0.094 | 0.462 | 0.732 | NS | 116: | *FCER1A* | Mast Cell | 3.899 | 4.448 | 0.748 | 0.784 | 0.578 | 0.321 | 0.496 | NS | 117: | *MS4A2* | Mast Cell | 0.719 | 0.788 | 0.926 | 0.076 | **2.9×10^{-3}** | 0.321 | 0.325 | Model-sensitive (1/5) | 118: | *CPA3* | Mast Cell | 1.583 | 1.587 | 0.992 | 0.825 | 0.755 | 0.858 | 0.979 | NS | 119: 120: *Notes: Bold indicates $q < 0.05$. NPY was pre-specified but absent from the expression matrix. FDR correction applied to 16 measured genes per model. Female-only analysis: 91 FM / 41 HC.* 121: 122: **Key observations:** 123: - *MDGA2* and *DRD2* survive all five models including sex-adjusted OLS, suggesting that their signal is not solely explained by the severe sex imbalance in the cohort. 124: - *CAMKV* and *CELF4* survive four models (including female-only) but not the sex-adjusted OLS on the full cohort ($q = 0.112$ and 0.167 , respectively). Their signals are supportive but model-sensitive. **Crucially, the OLS model ($\log_2(\text{FPKM}+1) \sim \text{case} + \text{sex}$) suffers from severe multicollinearity because fibromyalgia cases and female sex are highly collinear (95% of cases are female while controls are evenly split). This collinearity dramatically inflates the standard errors of the coefficients, leading to a profound loss of statistical power for case status. The female-only subgroup analysis (91 FM vs. 41 HC) completely removes the sex variable, thus eliminating this multicollinearity confound and rescuing the significance of both *CAMKV* and *CELF4* ($q = 0.032$). This highlights the female-only subgroup as the primary, statistically unconfounded model.** 125: - *HTT* — the gene with the strongest GWAS coding variant — shows a supported signal (3/5 models), which was not apparent in the original FPKM-only analysis. However, we note that the direction of effect is negative (downregulated in FM PBMCs), contrasting with the upregulation seen in *MDGA2* and *DRD2*. 126: - No mast cell marker shows a robust PBMC signal; *MS4A2* reaches significance only in the Mann-Whitney model and is therefore classified as model-sensitive rather than replicated across analytical frameworks. 127: 128: ### 3.2 The Opioid/Tachykinin Neuropeptide Axis Is Elevated in FM PBMCs — but the Signal Is Compositional 129: 130:

Extending the panel to the neuropeptide axis (validate_fm_biomarkers_iter2.py v3 + session analysis), we find that **ligands and receptors of both the tachykinin (Substance P) and endogenous opioid systems are simultaneously elevated** in FM PBMCs (Table 2). This is the largest effect-size block observed in the entire investigation, and it survives all five sex-sensitivity models (§2.4). It does **not**, however, survive adjustment for cell composition — the analysis that distinguishes per-cell transcriptional regulation from a shift in which cells are circulating. That result is presented below and is decisive for how the axis should be interpreted.

131: 132: **Table 2. Opioid/tachykinin axis in GSE221921 (96 FM vs 93 HC PBMCs).**

133: 134: | Gene | Role | FC arith. | FC geom. | MWU p | Bonf (×21) | d (FPKM) | d (log₂) | Composition-adj. p | Classification |

135: | — | — | — | — | — | — | — | — | — | — |

136: | **TACR1** | NK1 receptor (Substance P) | **2.73** | 1.21 | **0.0010** | 0.021 | **+0.60** | +0.64 | 0.039-0.117 | **significant, medium effect; compositional** |

137: | **OPRM1** | μ-opioid receptor | **2.28** | 1.61 | **<0.0001** | <0.0021 | **+0.53** | +0.72 | 0.030-0.080 | **significant, medium effect; compositional** |

138: | **TAC1** | Substance P (ligand) | **2.10** | 1.31 | **0.0002** | 0.0042 | **+0.47** | +0.56 | 0.080-0.393 | **significant, small-medium; compositional** |

139: | **OPRK1** | κ-opioid receptor | **1.78** | 1.13 | **0.0017** | 0.036 | **+0.38** | +0.43 | 0.796-0.836 | **significant, small-medium; compositional** |

140: | **PENK** | Enkephalins (ligand) | 1.38 | 1.33 | 0.0031 | 0.065 (marginal) | +0.21 | +0.36 | 0.250-0.305 | trend (Bonf ×19 = 0.060) |

141: | **OPRD1** | δ-opioid receptor | 1.23 | 1.04 | 0.0199 | NS | +0.10 | +0.16 | 0.655-0.930 | trend |

142: | **PNOC** | Nociceptin | 0.96 | 0.96 | NS | — | -0.02 | -0.08 | — | flat |

143: | **POMC** | β-endorphin/ACTH | 1.04 | 1.09 | NS | — | +0.03 | +0.13 | 0.917-0.928 | flat |

144: 145: *Bonf ×21 = conservative correction across the 21-gene union panel (19 + TACR1 + OPRK1). PENK is marginal under Bonf ×19 (p = 0.060) and does not survive ×21.*

146: *FC arith. = ratio of arithmetic group means (the value reported in earlier versions); FC geom. = ratio of geometric means, i.e. 2^{^(mean log₂(FPKM+1) difference)}. The two diverge because FPKM distributions are heavy-tailed (Shapiro-Wilk rejects normality for every gene, both groups; §2.3); the geometric ratio better describes the typical sample. Cohen's d is reported on both scales and is unchanged or larger under log₂, confirming the effects are not artifacts of outliers.*

147: *Composition-adj. p = p-value for case status in log₂(FPKM+1) ~ case + sex + cell fractions (model 6, §2.4); the range spans four deconvolution implementations. No axis gene is significant across all four (§3.2, analysis/E1b_variant_sweep.csv).*

148: 149: **Co-expression of the axis (Spearman, all 189 samples):**

150: 151: | Pair | rho | p | 152: | — | — | — | 153: | OPRM1 ↔ TAC1 | +0.632 | <0.0001 | 154: | PENK ↔ OPRM1 | +0.442 | <0.0001 | 155: | PENK ↔ TAC1 | +0.408 | <0.0001 | 156: | TAC1 ↔ TACR1 | +0.382 | <0.0001 | 157: | PENK ↔ POMC | +0.312 | <0.0001 | 158: 159: **Cell-composition adjustment (model 6; scripts/e1 deconvolution adjusted model.py).** Bulk

PBMC expression confounds transcription per cell with the mix of cells present. Regressing each axis gene on case + sex + 12 estimated leukocyte fractions collapses the signal: **no axis gene is significant across all four deconvolution implementations tested** (Table 2, “Composition-adj. p”). OPRM1 and TACR1 are borderline (significant under marker-score fractions, $p = 0.030$ and 0.039 ; not under NNLS fractions, $p = 0.080$ and 0.117); TAC1, PENK, OPRK1, OPRD1 and POMC fail under all four ($p = 0.25$ – 0.93). 160: 161: To establish that this adjustment is a genuine filter rather than a procedure that removes all signal, the same model was applied to 600 randomly selected expressed genes with cell-type markers excluded. Of the 273 with a case effect after sex adjustment, **76 (28%) remained significant after composition adjustment** (median $p = 0.182$). The adjustment removes roughly three-quarters of case effects, but a substantial minority survives — so failing it is informative, and so is passing it. Collinearity among the fractions is moderate (VIF median 2.7, max 13.1). 162: 163: **Interpretation.** The axis is a **coordinated block**: ligands (*TAC1*, *PENK*) and their cognate receptors (*TACR1*, *OPRM1*, *OPRK1*) are jointly elevated and positively co-expressed, with *TACR1* (NK1 receptor for Substance P; $d = +0.60$) exceeding its ligand in effect size. But the composition analysis reframes what that block means: the elevation is **attributable to a shift in circulating leukocyte populations, not to upregulation within a given cell type**. This is a weaker claim than “the SP→NK1 circuit is sensitized,” and we do not make the stronger one. It remains a real and potentially informative peripheral phenotype — a change in which neuropeptide-expressing cells circulate in FM — but it is a compositional phenotype, and the appropriate follow-up is cell-resolved (flow cytometry or scRNA-seq), not bulk. Two further confounders cannot be separated with these data: chronic opioid exposure is common in FM and is known to regulate opioid receptor expression (Limitation 10), and it may itself alter leukocyte composition. Elevated *OPRM1/OPRK1* therefore cannot be read as compensatory activation of the endogenous opioid system on this evidence. The possible link to pro-inflammatory cytokine involvement reviewed by Rodríguez-Pintó et al. (2014) via mast cell/neutrophil trafficking is not established in that review and remains hypothetical. 164: 165: ### 3.3 Cross-Context Comparison: GSE67311 (Whole Blood) 166: 167: To assess whether these signals are detectable in a different cell fraction, we examined the same gene sets in GSE67311 (whole blood, Affymetrix microarray; Table 3). 168: 169: **Table 3. Cross-context comparison: GSE221921 (PBMCs) vs. GSE67311 (whole blood).** 170: 171: | Gene | Category | GSE221921 (PBMCs) | GSE67311 (whole blood) | Replicates? | 172: |——| |——| |——| |——| |——| 173: | *CPA3* | Mast Cell | NS | 1.8×10^{-6} | Whole blood only | 174: | *MS4A2* | Mast Cell | Model-sensitive | 1.5×10^{-5} | Whole blood only | 175: | *FCER1A* | Mast Cell | NS | 1.7×10^{-5} | Whole blood only | 176: | *HDC* | Mast Cell | NS | 4.4×10^{-5} | Whole blood only | 177: | *TAC1* | Tachykinin

ligand | FC=2.10, p=0.0002 | FC=1.004, p=0.495 | NO | 178: | *OPRM1* | Opioid receptor | FC=2.28, p<0.0001 | FC=1.020, p=0.160 | NO | 179: | *IL6* | Inflammatory | FC=1.66, p=0.0002 | FC=1.023, p=0.524 | NO | 180: | *PENK* | Opioid ligand | FC=1.38, p=0.0031 | FC=1.038, p=0.031, Bonf=0.248 | trend only | 181: | *PCSK1N* | Prohormone convertase | FC=0.76 (↓) | FC=1.041 (↑), p=0.045 | INVERTED | 182: | *MDGA2* | GWAS Neural | Robust (5/5) | 0.99 | PBMCs only | 183: | *DRD2* | GWAS Neural | Robust (5/5) | 0.42 | PBMCs only | 184: 185: No gene was significant in both datasets. The PBMC-derived neuropeptide and GWAS-neural signals do not replicate in whole blood; the mast cell/basophil panel is detectable only in whole blood.

Deconvolution analysis (2026-08-03) — using cell-type marker averages for neutrophils, T cells, B cells, monocytes, and NK cells in GSE67311 — shows **essentially identical cellular composition between FM and HC** (neutrophil score: FM = 10.92 vs HC = 10.89, 7 markers; lymphocytes/monocytes/NK all within 0.1), and weak/mixed correlation of the neutrophil score with the PBMC proxies (TAC1 r = +0.07 NS; OPRM1 r = -0.22, p = 0.008; PENK r = -0.13 NS). **The non-replication is therefore NOT explained by neutrophil-driven dilution of the whole-blood signal.** Remaining explanations are platform differences (RNA-seq FPKM vs. microarray RMA), cohort heterogeneity, small true effects (d ≈ 0.2-0.5) that do not survive inter-platform noise, or partial cohort-specificity of the GSE221921 findings. 186: 187:

Opioid axis extension (2026-08-03, scripts/validate_opioid_axis_gse67311.py). The prior analysis covered TAC1/OPRM1/IL6 only. We extended it to the full opioid/tachykinin receptor set (TACR1, OPRM1, OPRK1) plus TAC1 and PENK (Mann-Whitney + Bonferroni ×5 + Cohen's d). **Absolute**

expression does not replicate (TACR1 FC = 1.008, p = 0.480; OPRM1 FC = 1.020, p = 0.160; OPRK1 FC = 1.021, p = 0.878; TAC1 FC = 1.004, p = 0.495; PENK FC = 1.038, p = 0.031, Bonf = 0.155 — none survives). 188: 189: **Co-expression architecture (Spearman, all 10 pairs, FM-pooled vs HC-pooled):** 190: 191: | Pair | ρ_FM | p_FM | ρ_HC | p_HC | FM > HC? | Pre-registered? | 192: | — | — | — | — | — | — | — | 193: | TACR1-OPRK1 | +0.736 | <0.001 | +0.405 | <0.001 | ✓ | yes | 194: | OPRM1-OPRK1 | +0.530 | <0.001 | +0.279 | 0.015 | ✓ | yes | 195: | TACR1-OPRM1 | +0.420 | <0.001 | +0.280 | 0.015 | ✓ | yes | 196: | OPRM1-TAC1 | +0.363 | 0.003 | +0.078 | 0.504 | ✓ | yes | 197: | OPRK1-PENK | +0.375 | 0.002 | +0.150 | 0.200 | ✓ | yes | 198: | TACR1-TAC1 | +0.189 | 0.125 | +0.420 | <0.001 | **x HC > FM | no — omitted** | 199: | OPRK1-TAC1 | +0.259 | 0.034 | +0.370 | 0.001 | **x HC > FM | no — omitted** | 200: | TACR1-PENK | +0.257 | 0.036 | +0.029 | 0.806 | ✓ | **no — omitted** | 201: | OPRM1-PENK | +0.200 | 0.104 | +0.201 | 0.084 | ≈ | **no — omitted** | 202: | TAC1-PENK | -0.003 | 0.982 | +0.076 | 0.518 | ≈ | **no — omitted** | 203: 204: **Correction note (2026-08-04).** A prior version of this section reported only 5 of 10 co-expression pairs, selected by direction (FM > HC). The 5 omitted pairs include 2 where HC actually exceeds FM (TACR1-TAC1, OPRK1-

TAC1), which contradicts the earlier statement “HC pairs consistently weaker or non-significant.” That statement is corrected here: **HC co-expression is weaker for 5 of 10 pairs, but comparable or stronger for 3, and near-null for 2.** The 5 originally reported pairs are those where the FM co-expression exceeds HC; reporting them without the full 10-pair table constituted selective reporting. A formal Fisher r-to-z test of the FM-vs-HC difference for each pair should accompany any inferential claim. The circuit-level interpretation (transcriptional coherence of the opioid/tachykinin axis in FM) rests on the 5 pairs where FM > HC, but is not a universal property of the 10-pair matrix. 205: 206: **Interpretation:** the opioid/tachykinin circuit shows *some* transcriptional coherence in FM whole blood, but **only 1 of 10 pairs differs significantly between FM and HC** (TACR1-OPRK1, Fisher r-to-z p = 0.003). The remaining 9 pairs do not distinguish the groups. The interpretation of a “coherent transcriptional module” is **not supported by the data** as a universal property: 2 pairs reverse (HC > FM), and the rest include near-null values. Its absolute amplitude is compartment-dependent (PBMC-specific). Plasma protein measurement (Olink/ELISA), not whole-blood transcriptomics, remains the decisive validation compartment. **Technical caveat:** in whole blood (GSE67311) TAC1 sits at the 8th intensity percentile, OPRK1 at p13, and OPRM1 at p19 of the microarray distribution (vs ACTB at p99). Correlations among probes in the bottom decile can reflect shared background noise rather than genuine co-regulation — the significant pair should be interpreted with this limitation. 207: 208: **Honest assessment.** These PBMC-derived proxies are **PBMC/RNA-seq-specific** and are **not replicated in whole blood**. They should be described as “PBMC-specific transcriptional signatures,” not as validated peripheral blood biomarkers. The planned plasma Olink/ELISA validation (§7) measures the relevant compartment (plasma protein) and is the definitive test; the whole-blood non-replication qualifies but does not invalidate the plasma hypothesis. 209: 210: ### 3.4 UK Biobank Causal Proteomics: CA14 mRNA Elevated in PBMcs (Plasma ↓), Immune-Signaling Genes Downregulated in FM PBMcs 211: 212: To connect population-scale causal proteomics with our transcriptomic data, we tested the UKB causal genes (Li ZY et al., 2025; Chen et al., 2025) in GSE221921. Eight of the causal genes are measurable in the matrix (LEP and TNF absent). 213: 214: **Table 4. UKB causal genes for chronic pain / CWP in FM PBMcs (GSE221921).** 215: 216: | Gene | UKB source | FC arith. | FC geom. | MWU p | Bonf ×9 | d (FPKM) | d (log₂) | Direction | 217: |——|———|———:|———:|———:|———:|———:|———:|———:| 218: | **CA14** | MR+coloc causal CWP (top-ranking) | **2.29** | 1.32 | **0.0003** | **0.0027** | **+0.41** | +0.56 | ↑↑ **significant (sex-confounded, §4.2)** | 219: | TNFRSF1B | MR causal chronic pain — **back-specific** (Li ZY 2025) | 0.541 | 0.45 | <0.0001 | <0.0001 | -0.56 | -0.84 | ↓↓ significant | 220: | CD74 | MR causal chronic pain — **hip-specific** (Li ZY 2025) | 0.579 | 0.40 | <0.0001 | <0.0001 | -0.44 | -0.80 | ↓↓ significant | 221: |

COL18A1 | MR causal chronic pain — **abdominal-specific** (Li ZY 2025) | 0.581 | 0.64 | 0.0001 | 0.0005 | -0.54 | -0.58 | ↓↓ significant | 222: | BTN2A1 | MR causal chronic pain — **knee/abdominal-specific** (Li ZY 2025) | 0.745 | 0.72 | 0.0025 | 0.023 | -0.35 | -0.46 | ↓ significant | 223: | TNFRSF4 | MR causal chronic pain — **knee-specific** (Li ZY 2025) | 0.768 | 0.74 | 0.0004 | 0.0036 | -0.23 | -0.44 | ↓ significant | 224: | CD302 | MR causal chronic pain | 0.717 | 0.85 | 0.154 | NS | -0.19 | -0.22 | ↓ trend | 225: | TNFRSF9 | MR causal chronic pain | 1.459 | 1.07 | 0.805 | NS | +0.27 | +0.17 | ~flat | 226: 227: *FC columns as defined for Table 2. TNFRSF9 illustrates why both are reported: an arithmetic ratio of 1.46 corresponds to a geometric ratio of 1.07 and a rank-based p of 0.805 — the arithmetic ratio is driven by a small number of high-FPKM samples, not by a shift in the bulk of the distribution.* 228: 229: **CA14: causal in plasma (downregulated), elevated in PBMC mRNA (unadjusted) — a direction-sensitive triangle with a sex-confound caveat.** CA14 is the **top-ranking causal protein for chronic widespread pain** (MR + colocalization PP.H4 > 0.5; Chen et al., 2025) and the **only UKB causal gene that is upregulated in FM PBMC mRNA in the unadjusted analysis** (FC = 2.29, p = 0.0003, d = +0.41). **Sex confound caveat (2026-08-04):** the female-only primary model yields p = 0.135, d = +0.24, FC = 1.60 — the unadjusted signal does not survive sex stratification (see §4.2). Crucially, the **published plasma direction is the opposite:** in the UKB cross-sectional analysis, CA14 is among the **ten most downregulated plasma proteins** in CWP, while MR indicates that **genetically elevated CA14 is protective** (discordant observational vs. MR direction; Chen et al., 2025). The authors therefore propose **CA14 agonists, rather than the inhibitor sulthiame (ChEMBL328560)**, as the more promising therapeutic direction, given non-linear CA14-pain associations. Interpretation: low plasma CA14 may contribute causally to pain (consistent with pH/nociception dysregulation); the PBMC mRNA elevation we observe may be compensatory or compartment-specific. This makes CA14 the candidate with the strongest *cross-level* support (plasma proteomics + MR + colocalization + PBMC mRNA), but not the strongest candidate in this dataset: its PBMC signal does not survive sex stratification (3/5 models), and COL9A1/PTN outrank it on every within-dataset criterion (§3.4.5). CA14 is therefore carried forward as a **direction-specific reference** with a testable prediction — **CA14 should be ↓ in FM plasma** in the Olink validation — and a repurposing direction of agonism, not sulthiame inhibition. 230: 231: **Five of nine UKB causal genes are significantly DOWNREGULATED in FM PBMCs** (TNFRSF1B, CD74, COL18A1, BTN2A1, TNFRSF4; FC 0.54-0.77). These genes encode TNF receptors and immune signaling molecules. **Important caveat:** these MR hits are **site-specific** (back, hip, abdominal, knee) in Li ZY 2025 — not CWP/FM-specific — and none of the five is among the 18 causal CWP proteins identified in Chen 2025. The plasma direction of these

five proteins in FM is not known (the UKB cross-sectional analysis reported by Chen 2025 does not cover them specifically). The coordinated downregulation in PBMC mRNA is therefore consistent with — but not proof of — an **immunomodulatory/exhaustion hypothesis** (reduced TNF-family and immune signaling), which remains explicitly speculative pending plasma measurement. 232: 233: ### 3.4.5 COL9A1-PTN: The Bonferroni-Surviving Extracellular-Matrix / Neurite-Outgrowth Module 234: 235: Having established that CA14 does not survive sex stratification (§4.2), we asked which genes from the chronic-widespread-pain causal set (Chen 2025) represent the most robust transcriptomic signal in FM PBMCs. Testing the 18 CWP causal proteins in GSE221921 under the five-model framework (E2), **only two genes survive Bonferroni correction on the sex-adjusted model**: *COL9A1* (FC = 2.32, d = +0.88, p_adj_Bonf = 8.5×10^{-5}) and *PTN* (FC = 2.91, d = +0.61, p_adj_Bonf = 0.020). *BPIFB2* is nominally significant (p_adj = 0.0035) but does not survive Bonferroni (p_adj_Bonf = 0.063); *ST3GAL1* is downregulated (FC = 0.765) and anticorrelated with the pair (r = -0.23). Under the five-model framework defined in §2.4, both *COL9A1* and *PTN* are **Robust 5/5** (*COL9A1* p = 1.1×10^{-8} to 2.3×10^{-5} ; *PTN* p = 1.4×10^{-6} to 9.6×10^{-5} across models 1-5). An earlier version described them as “Robust 4/5”; that score was computed against the alternative model set in *E7_sensitivity_master_GSE221921.csv*, in which a male-only stratum of **5 FM patients** replaces model 2 and which every gene in the panel fails for lack of power. See §2.4 for why that stratum is not used as a robustness criterion. 236: 237: Geometric-mean fold changes are smaller than the arithmetic ratios quoted above (*COL9A1* 2.32 → 1.72; *PTN* 2.91 → 1.33), while standardized effects are stable or larger under log₂ (*COL9A1* d = +0.88 → +0.86; *PTN* d = +0.61 → +0.72). 238: 239: **Cell-composition adjustment (model 6) — the discriminating test.** Applying the composition-adjusted model that collapses the neuropeptide axis (§3.2), **COL9A1 and PTN are the only genes in this investigation that survive**, and they do so under all four deconvolution implementations: *COL9A1* p = 0.012-0.020 (β = +0.370), *PTN* p = 0.028-0.046 (β = +0.223). *PTN* is marginal under NNLS-estimated fractions (p = 0.046) and we report it as such. Against the negative-control benchmark — 28% of random case-associated genes survive this adjustment (§3.2) — passing it is meaningful but not extraordinary; the strength of the *COL9A1*-*PTN* result lies in surviving sex stratification, Bonferroni correction **and** composition adjustment simultaneously, which no other signal in the panel does. 240: 241: **Co-expression (C1).** *COL9A1*, *BPIFB2* and *PTN* form a positively co-expressed triad (r = 0.25-0.51, all p < 0.001); *ST3GAL1* anticorrelates weakly (r = -0.15 to -0.24). Fisher r-to-z shows the correlations do **not** differ between FM and HC (p > 0.17 for all pairs) — the module is a structural property of the co-expression network, not a disease-specific rewiring. 242: 243: **Biological function (C3).** The triad resolves into two convergent axes rather than a single module: (i) a **structural/neural axis** —

COL9A1 (collagen alpha-1(IX), minor fibrillar cartilage collagen; mutations cause multiple epiphyseal dysplasia and osteoarthritis, a direct joint-pain link) co-expressed with *PTN* (pleiotrophin, secreted growth factor driving neurite outgrowth and nerve repair); (ii) an **innate-immunity axis** — *BPIFB2* (LPS-binding lipid transfer, Sjögren biomarker) with *ST3GAL1* (T-cell sialylation, BDNF sialylation). The *COL9A1*-*PTN* axis is the stronger and more novel: both are secreted and detectable in plasma Olink, and both connect to pain through anatomically distinct but clinically central routes (joint integrity and nociceptive nerve plasticity).

244: 245: **Power analysis for plasma validation (C4).** Anchoring the planned Olink cohort to the observed mRNA effect ($d = 0.88$, $FC = 2.32$ for *COL9A1*) and modeling mRNA→protein attenuation at four scenarios ($r = 0.8 / 0.6 / 0.4 / 0.3$), the required sample size for 80% power is: 33/group (optimistic), 56/group (moderate), 129/group (conservative), 233/group (pessimistic). A cohort of **75 FM + 75 HC** covers the moderate scenario with 88% power and is the recommended design.

246: 247: **Interpretation.** *COL9A1* and *PTN* replace *CA14* as the **primary plasma-validation candidates** from the UKB causal set: they are Bonferroni-surviving, sex-adjusted-robust, plasma-detectable, and biologically linked to pain through joint and neural mechanisms. *CA14* is retained as a direction-specific **reference** candidate (expecting ↓ in FM plasma, per Chen 2025), not as the lead target.

248: 249: **### 3.5 Targeted Literature Review: Dopamine Agonists in FM**

250: 251: Our literature search identified three clinical studies and one preclinical study (Table 5):

252: 253: **Table 5. Dopamine agonist studies in fibromyalgia.**

254: 255: | Study | Drug | Type | N | Result | Key Finding | Risk of Bias |

256: | --- | --- | --- | --- | --- | --- | --- |

257: | Holman & Myers, 2005 (PMID 16052595) | Pramipexole | RCT (DB-PC) | 60 | **Positive** | 36% pain ↓ vs 9% placebo; 42% achieved ≥50% pain decrease | High: single-center, author held patents on D2/D3 use in FM |

258: | Holman, 2003 (ACR conference) | Ropinirole | Pilot | 30 | NS ($p=0.31$) | Underpowered. Not published in peer-reviewed journal. | Very high: unpublished, tiny N |

259: | GSK NCT00256893 | Ropinirole CR | Phase II RCT | 160 | **Negative** | Failed primary endpoints | Moderate: sponsor-reported, results not published in peer-reviewed journal |

260: 261: **### 3.6 GSE269047 — excluded**

262: 263: GSE269047 was evaluated as a replication cohort but was found to contain primarily HERV (human endogenous retrovirus) transcripts, not annotated gene expression; none of our target genes are measurable on that platform. The accession suffixes (opti/bgrd/rand) are probe design categories, not gene symbols, and the cohort is a mixed FM/ME-CFS sample. The dataset is therefore **not usable as a proxy-validation dataset** and we draw no conclusions from it (GSE269047_NO_UTILIZABLE.md).

264: 265: — 266: 267: **## 4. Discussion**

268: 269: **### 4.1 The Peripheral Signature of FM Is Neuropeptide/Nociceptive and Immunomodulatory, Not Classically Inflammatory**

270: 271: The central finding of this study is that the peripheral transcriptomic

signature of FM — as measured in PBMCs (GSE221921) — is dominated by three non-inflammatory axes: 272: 273: 1. **An extracellular-matrix / neurite-outgrowth module is upregulated cell-intrinsically** — *COL9A1* (FC = 2.32, d = +0.88) and *PTN* (FC = 2.91, d = +0.61) (§3.4.5). This is **the most robust finding of the investigation**: it is the only signal that survives sex stratification, Bonferroni correction on the sex-adjusted model, **and** adjustment for cell composition, simultaneously and across all four deconvolution implementations tested. Both proteins are secreted and plasma-detectable (Olink), and connect to pain through joint integrity (*COL9A1*, osteoarthritis link) and nociceptive nerve plasticity (*PTN*, neurite outgrowth). This is the primary plasma-validation target of the investigation. 274: 2. **A complete opioid/tachykinin neuropeptide block is elevated, but compositionally** (*TACR1*, *OPRM1*, *TAC1*, *OPRK1*; d = +0.38 to +0.60, co-expressed rho 0.31-0.63). This is the largest effect-size block in the investigation and survives all five sex-sensitivity models, but **no axis gene survives cell-composition adjustment across implementations** (§3.2). The signal therefore reflects a shift in which neuropeptide-expressing leukocytes circulate in FM, not upregulation within a cell type — a real peripheral phenotype, but a different claim from circuit sensitization, and one that chronic opioid exposure could equally produce (Limitation 10). 275: 3. **CA14 — the top UKB causal protein for chronic widespread pain — is elevated in PBMC mRNA in the unadjusted analysis** (FC = 2.29) **but does not survive the female-only primary model** (p = 0.135, FC = 1.60; sex-confounded) **and is downregulated in plasma** in the UKB cross-sectional analysis, with MR indicating that genetically elevated CA14 is protective; five UKB causal immune-signaling genes are downregulated (*TNFRSF1B*, *CD74*, *COL18A1*, *BTN2A1*, *TNFRSF4*). CA14 is retained as a direction-specific reference candidate (expecting ↓ in FM plasma), not as the lead target. 276: 277: At the same time, the classical inflammatory axis (IL-6/IL-8) — the historical favorite in FM biomarker research — receives **no support at population scale**: IL-6 and IL-8/CXCL8 do not figure among the highlighted proteins in the largest published plasma proteomics screen of chronic pain (51,644 UKB participants, 2,923 proteins; Li ZY et al., 2025), though this is **weak evidence** — the 474 pain-associated proteins reside in supplementary tables that were not consulted, so absence from the narrative text does not constitute absence of association. **TAC1 (Substance P) is not covered by the Olink Explore panel and therefore cannot be evaluated with these data — its absence from the text is not informative.** In our own PBMC data, IL6 is significantly elevated (FC = 1.66, p = 0.0002) but with a **small effect size** (d = +0.31), and CXCL8 is inverted in PBMCs (down relative to whole blood, consistent with cell-fraction biology). 278: 279: The coherent narrative that emerges is **extracellular-matrix / neurite-outgrowth dysregulation (COL9A1-PTN) as the cell-**

intrinsic core + a compositional shift in neuropeptide-expressing leukocytes + immunomodulation (UKB immune genes) + pH/nociception dysregulation (CA14), not systemic inflammation: reduced TNF-family signaling (5 UKB-causal genes ↓), an extracellular-matrix/nerve-repair module upregulated within cells (COL9A1-PTN), a compositional enrichment of cells bearing the nociceptive neuropeptide programme (SP→NK1, opioid receptors), and a pH-regulating carbonic anhydrase (CA14) that is causal and **downregulated in plasma** (elevated in PBMC mRNA). This reframing has direct consequences for biomarker selection: the plasma validation panel should prioritize **COL9A1 and PTN** (Olink — primary targets, Bonferroni-surviving, plasma-detectable), with CA14 measured as a direction-specific reference (expecting ↓ in FM plasma) and Substance P/enkephalins (ELISA) as secondary neuropeptide measures; IL-8 is retained only as a technical assay control. 280: 281: ### 4.2 CA14: A Cross-Level Causal Candidate — Downregulated in Plasma, Druggable by Agonism 282: 283: CA14 emerges as the most actionable candidate in this investigation, with a direction-sensitive evidence stack: 284: 285: | Layer | Evidence | Source | 286: |——|———|———| 287: | 1. Plasma protein | Causal in CWP (top-ranking, MR + colocalization PPH4 > 0.5); **downregulated (↓) cross-sectionally**; MR protective for genetically elevated CA14 | Chen et al. 2025, *Adv Sci* (PMID 41025730), 29,254 participants | 288: | 2. PBMC mRNA (unadjusted) | ↑ in FM (FC = 2.29, p = 0.0003, d = +0.41) | GSE221921 (this study, full cohort) | 289: | 2b. PBMC mRNA (female-only) | **Does not survive** (p = 0.135, d = +0.24, FC = 1.60) | GSE221921, female-only subgroup (91 FM / 41 HC) | 290: | 3. Pharmacology | Sulthiame (ChEMBL328560) is an **inhibitor; the paper proposes CA14 agonists — not inhibitors — as the more promising therapeutic direction** | Chen et al. 2025; ChEMBL | 291: 292: **Sex confound correction (2026-08-04)**. The unadjusted CA14 signal (FC = 2.29, p = 0.0003) does not survive the sex-stratified sensitivity analysis that §2.4 defines as the “primary, statistically unconfounded model” — the female-only subgroup (91 FM / 41 HC) yields p = 0.135, d = +0.24, FC = 1.60. Diagnostic: within healthy controls, CA14 expression in females is 2.41× that of males (mean F = 0.901, mean M = 0.374, p = 0.034), and the FM cohort is 95% female while the HC cohort is 55% male. The unadjusted FC = 2.29 is therefore confounded by the structural sex imbalance — it is of the same magnitude as the pure sex effect (F/M ratio = 2.41). **CA14 should be reported as “sex-confounded, not surviving the primary model” in the PBMC mRNA layer.** The plasma protein layer (Layer 1, Chen et al. 2025) is unaffected by this confound because the UKB analysis was sex-adjusted. The testable prediction (↓ CA14 in FM plasma) remains valid and is independent of the PBMC mRNA result. 293: 294: CA14 encodes carbonic anhydrase XIV, a membrane-bound enzyme regulating extracellular pH; pH dysregulation in nociceptors is a well-established driver of pain signaling (acid-sensing). The published analysis shows CA14 among the ten most downregulated plasma

proteins in CWP, while MR indicates a protective effect of genetically elevated CA14 — an observational-vs-MR discordance the authors interpret as state-dependent protein alteration vs. lifelong genetic predisposition. **Our testable prediction: CA14 should be ↓ in FM plasma** in the Olink validation; if confirmed, the repurposing direction is **agonism/activation of CA14**, not sulthiame inhibition. The elevated PBMC mRNA we observe may be compensatory or compartment-specific; both compartments should be measured to resolve this. **Mechanistic caveat (QSP, 2026-08-04, thermodynamically corrected):** a single-compartment QSP model of the CA14→pH→ASIC pathway, recalibrated so the uncatalyzed rates respect the thermodynamic Keq ($K_{\text{UNCAT_R}} = K_{\text{UNCAT_F}} / K_{\text{eq}} = 189.3 \text{ s}^{-1}$, correcting a $3.8\times$ bug), shows $\Delta\text{pH} = 0$ exactly — because at steady state the CA-catalyzed terms cancel identically ($k_{\text{buf}} \cdot h = J_{\text{co2}} + J_{\text{acid}} - k_{\text{diff}} \cdot c$), so the enzyme changes the relaxation rate, not the equilibrium. The model's basal pH is set by $J_{\text{co2}}/k_{\text{buf}}$ and does not reproduce the Henderson-Hasselbalch value of 7.33 across any parameter combination (it yields 7.83/7.21/6.76 depending on unmeasured fluxes), indicating the pH baseline is not properly calibrated. **The simple peripheral acidosis mechanism is therefore not evaluable with this model — a model that cannot exhibit the effect cannot refute it.** The earlier conclusion (“vía descartada”) is retracted; the correct framing is “hipótesis no evaluable con este modelo de estado estacionario de compartimento único.” The causal signal (MR/coloc from Chen 2025) remains statistically valid and may act via a different compartment (CNS), via transient pH kinetics, or reflect CA14 as a marker rather than mediator (see scripts/qsp_ca14_ph_nociception.py). 295: 296: **Druggability caveat.** Sulthiame inhibits multiple carbonic anhydrases, not only CA14, and the published analysis explicitly favors agonists over inhibitors for pain; no well-documented CA14 activator is yet available. Any repurposing hypothesis requires isoform selectivity assessment and cannot be inferred from the present transcriptomic analysis alone. 297: 298: ### 4.3 The GWAS-Transcriptomic Convergence on Neural Genes 299: 300: The GWAS-prioritized neural genes *MDGA2* and *DRD2* show robust upregulation across multiple analytical models in FM PBMCs, including sex-adjusted and female-only sensitivity analyses. Two additional genes (*CAMKV*, *CELF4*) show supportive but sex-covariate-sensitive signals. This convergence of genetic risk (GWAS) and transcriptomic alteration (independent cohort) across independent methodologies is suggestive of biological relevance, though it remains an exploratory observation. 301: 302: We emphasize that the GWAS network is not exclusively “dopaminergic.” *MDGA2* encodes a GPI-anchored immunoglobulin superfamily member involved in synaptogenesis and neural circuit formation. *CAMKV* is a CaM kinase-like protein involved in dendritic spine dynamics. *CELF4* regulates neuronal mRNA metabolism. Only *DRD2* is strictly dopaminergic. The finding is therefore better characterized as convergence on a **neural/**

synaptic GWAS network that includes, but is not limited to, dopaminergic signaling. 303: 304: Our computational target profiling of MDGA2 (the most significant hit in our PBMC reanalysis, $q = 1.1 \times 10^{-7}$) using AlphaFold tridimensional models and clinical database mapping (Open Targets) provides additional mechanical insights. AlphaFold predicts a highly structured and ordered protein (Global pLDDT = 84.81), characterized by a large rigid 6-domain Ig-like supradomain separated by a flexible six-residue linker from the C-terminal MAM domain. This structural flexibility is crucial for modulating intercellular synaptogenesis. Furthermore, Open Targets database queries confirm that MDGA2 is highly constrained genetically (LoF score = 1.0, oe = 0.257), has direct clinical associations with chronic pain (Back Pain, score = 0.40), and is linked pharmacogenomically to the clinical outcomes of Milnacipran, an FDA-approved drug for Fibromyalgia. At the therapeutic level, its high-confidence extracellular GPI-anchored localization renders it highly tractable for antibody-based therapies or biologics targeting neuro-immune interactions.

305: 306: **### 4.4 The DRD2 Signal: Interpretation and Caveats**
307: 308: The upregulation of *DRD2* ($\text{Log}_2\text{FC} = +1.41$, $q = 2.9 \times 10^{-5}$; robust across all 5 models) in PBMCs warrants careful interpretation: 309: 310: **1. Absolute expression is low** (FM mean = 0.72 FPKM, HC mean = 0.27 FPKM). While the fold change and statistical significance are robust, the biological impact of sub-FPKM expression differences requires validation by targeted methods. Scientifically, an expression level under 1 FPKM in bulk tissue can represent either low-level “transcriptional noise” (leakage) across the bulk population or highly concentrated, biologically relevant expression restricted to a tiny immune subpopulation (e.g., specific $CD4^+$ or $CD8^+$ T cell subsets). To resolve this, orthogonal validation using highly specific qRT-PCR primers or single-cell qPCR is mandatory before drawing definitive functional conclusions. 311: 312: **2. Cell composition confounding.** *DRD2* is expressed in specific immune cells, such as T cell subsets, where it modulates cytokine production and chemotaxis (Pacheco et al., 2014). If FM patients have altered PBMC composition (e.g., different T cell subsets or monocyte proportions), the observed *DRD2* increase could reflect more cells expressing *DRD2* rather than per-cell upregulation. Without deconvolution analysis (CIBERSORTx, xCell, or similar), this cannot be distinguished. 313: 314: **3. Peripheral vs. central.** PBMCs are not the primary site of FM pathology. The GWAS heritability is enriched in brain tissues. Whether peripheral *DRD2* expression mirrors central dopaminergic dysfunction is unknown. 315: 316: **4. Independence from Cell-Type Abundances.** To explore whether the observed upregulation of *DRD2* is an artifact of altered PBMC proportions, we computed cell-type signature enrichment scores. *DRD2* expression did not correlate strongly with any estimated cell fraction ($r_{\text{max}} = 0.34$ with Tregs, and <0.30 with other fractions). This suggests that the *DRD2* signal in FM PBMCs represents genuine transcriptional upregulation

rather than a passive reflection of shifts in cellular composition, strengthening its biological validity. 317: 318: ### 4.5 Cell-Fraction-Dependent Contrast and the Whole-Blood Non-Replication 319: 320: The observation that mast cell markers are significant in whole blood but not PBMCs, while GWAS neural genes and the neuropeptide axis show the opposite pattern, is consistent with cell-fraction-dependent peripheral signatures. However, this observation cannot distinguish true cell-state changes from cell-composition differences, nor can it determine whether either signature is a disease driver versus a secondary biomarker. 321: 322: The non-replication of the neuropeptide proxies in GSE67311 is a **scientific negative that we report without cosmetic correction** (VALIDACION_GSE67311_NEGATIVA.md). Deconvolution rules out the most plausible mechanical explanation (neutrophil dilution). The remaining interpretations — platform, cohort, or true small effects — cannot be resolved with existing data. Practically, this means: (1) the proxies are PBMC/RNA-seq-specific; (2) the plasma Olink/ELISA protocol is the decisive test, because it measures the compartment (plasma protein) relevant to the causal UKB findings; (3) PCSK1N, which inverts direction between datasets, is re-classified as “not confirmed” until the discrepancy is understood. 323: 324: ### 4.6 Pharmacological Context: The 21-Year Gap and Druggability Barriers 325: 326: The pharmacological evidence for dopamine agonists in FM is limited. The sole positive RCT (Holman & Myers, 2005) carries substantial risk of bias (single-center, n=60, author held patents). The negative ropinirole trial (GSK NCT00256893) has never been published in a peer-reviewed journal, limiting independent evaluation. The current evidence is insufficient to recommend dopamine agonists for FM but does provide a rationale for re-examining this pharmacological axis in molecularly stratified cohorts. **Crucially, the “21-year gap” of non-replication is not merely an omission of research interest, but a reflection of the severe clinical tolerability barriers inherent to D2/D3 agonists in chronic pain populations. These ergot and non-ergot agonists are associated with severe side effects, including mesolimbic D3-receptor-mediated Impulse Control Disorders (ICDs) (e.g., pathological gambling, compulsive buying, hypersexuality), Dopamine Agonist Withdrawal Syndrome (DAWS) (characterized by profound anxiety, panic attacks, depression, and pain exacerbation upon tapering), orthostatic hypotension, and sudden “sleep attacks.” In a patient population already burdened by chronic fatigue, dysautonomia, and baseline sleep fragmentation, the therapeutic index for these compounds is extremely narrow, posing significant translation challenges.** 327: 328: The structural basis of this pharmacological axis — molecular docking of DRD2 with pramipexole and ropinirole, biophysical DRD2/DRD3 pocket mapping, de novo candidate design, and in silico validation across the fibromyalgia-implicated GPCR targets *DRD2*, *TACR1*, and

OPRM1 — is reported separately in an accompanying computational chemistry manuscript (Muñoz Rojas, companion paper: *Computational Docking and De Novo Design Against Fibromyalgia-Implicated GPCR Targets*), so that the present transcriptomic reanalysis remains focused on its primary evidence base. 329: 330: ### 4.7 Tissue-Invariant Germline Genomic sQTL Architecture of *DRD2* 331: 332: Given the non-replication and high volatility of peripheral blood mRNA measurements across independent cohorts (§3.3; §3.6 dataset excluded as non-annotated HERV platform), we emphasize that the strongest, methodologically unconfounded evidence supporting the *DRD2* axis in Fibromyalgia resides in **germline DNA genetics** (Tangente 3). 333: 334: Unlike blood mRNA expression levels — which fluctuate dynamically in response to cell-fraction shifts, acute physiological stress, and pharmacological therapies — germline genomic DNA variants are invariant across all tissues throughout an individual’s lifespan. The index GWAS risk SNP **rs2734833** (Kerrebijn et al., 2025) resides in strict linkage disequilibrium ($D' = 1.0$) with functional splicing quantitative trait loci (sQTLs), specifically **rs1076560** and **rs2283265**. 335: 336: GTEx v10 human brain tissue data (ENSG00000149295.14) demonstrate that baseline *DRD2* expression is highly concentrated in striatal structures (**Nucleus Accumbens: 54.21 TPM; Putamen: 46.80 TPM; Caudate: 41.40 TPM**), with moderate expression in Substantia Nigra (8.21 TPM) and low baseline levels (~ 0.66–1.22 TPM) across cerebral cortex and cervical spinal cord. In these central tissues, germline sQTL variants directly modulate the alternative splicing of **Exon 6** (87 bp encoding 29 amino acids: VVALSSQFPV SEAAEQARAE AQEAEEEVVG), which is inserted into the third intracellular loop (IL3). 337: 338: Exon 6 inclusion determines the functional ratio between two distinct receptor isoforms: 339: 1. ***DRD2*_{Short} (*D2S*)**: The presynaptic autoreceptor isoform (414 aa), which lacks Exon 6 and functions primarily to inhibit presynaptic dopamine synthesis and vesicular release via $G_{i\alpha2}$ coupling and Tyrosine Hydroxylase (TH) inhibition. 340: 2. ***DRD2*_{Long} (*D2L*)**: The postsynaptic signaling isoform (443 aa), which incorporates Exon 6 and mediates canonical postsynaptic dopaminergic neurotransmission via $G_{i\alpha1/3}$ and β -arrestin-2 / AKT-GSK3 β recruitment. 341: 342: Disruption of Exon 6 alternative splicing in central striatal and nociceptive circuits provides a cell-type-independent, tissue-invariant genomic mechanism through which genetic variation at the *DRD2* locus alters central pain processing, descending pain inhibition, and mesolimbic reward signaling in Fibromyalgia. Future experimental work should focus on germline sQTL genotyping and brain-isoform-specific quantification rather than relying on peripheral blood mRNA expression. 343: 344: — 345: 346: ## 5. Limitations 347: 348: 1. **Sex confounding**. GSE221921 has a severe sex imbalance (FM: 91F/5M; HC: 41F/52M). Our sensitivity analyses suggest that *MDGA2* and *DRD2* are

not solely explained by sex imbalance, while *CAMKV* and *CELF4* are sensitive to sex adjustment. Residual confounding remains possible, and future studies should use sex-balanced cohorts or sex-stratified designs. 349: 350: 2. **Statistical methodology.** FPKM with parametric tests is not gold standard for RNA-seq. Count-based modeling (DESeq2/edgeR) would be preferable, but raw counts were not available. The expanded panel uses Mann-Whitney U + Bonferroni + Cohen's d (post-adversarial-audit standard), which is robust to the FPKM distribution. 351: 352: 3. **Cell composition — the opioid axis is compositional; COL9A1/PTN are not (2026-08-04, revised).** Each gene was regressed against group + sex + 12 marker-estimated leukocyte fractions (model 6, §2.4). **No opioid-axis gene survives the adjustment across implementations** (OPRM1 $p = 0.030$ – 0.080 ; TACR1 $p = 0.039$ – 0.117 ; TAC1 $p = 0.080$ – 0.393 ; OPRK1, OPRD1, PENK, POMC $p = 0.25$ – 0.93). The elevated axis signal in FM PBMCs is therefore **compositional** — attributable to altered leukocyte fractions rather than transcriptional upregulation within a cell type. **COL9A1 and PTN, by contrast, survive under all four implementations** ($p = 0.012$ – 0.020 and 0.028 – 0.046), and are the only signals in the panel that do. A negative control (600 random expressed genes) shows the adjustment retains 28% of sex-adjusted case effects, so neither outcome is an artifact of the procedure (§3.2). 353: 354: Three caveats bound this analysis. (i) The fractions are **estimated from marker genes in the same expression matrix**, not measured by cytometry; adjusting for covariates derived from the same data risks over-adjustment, which the negative control bounds but does not eliminate. (ii) Marker sets for mast cells, basophils and dendritic cells share genes (CPA3, MS4A2, FCER1A, HDC), making the attribution of *which* population confounds a given gene unstable across implementations; the conclusion that composition confounds the axis does not depend on that attribution. (iii) Composition itself may be drug-induced — chronic opioid use is common in FM (Limitation 10) — so “compositional” does not distinguish disease biology from treatment. An earlier version of this limitation reported per-gene confounder assignments (NK cells for OPRM1, neutrophils for TACR1, etc.) from a single implementation; those assignments do not replicate across implementations and have been withdrawn. DRD2 was not formally deconvoluted and requires re-evaluation in the same framework. For GSE67311, marker-based deconvolution shows comparable composition between FM and HC, ruling out neutrophil dilution as the cause of non-replication. 355: 356: 4. **Low absolute expression.** DRD2 expression in PBMCs is < 1 FPKM. qRT-PCR validation is needed. 357: 358: 5. **Missing genes.** NPY was pre-specified but absent from the GSE221921 matrix (FDR applied to 16, not 17, genes). LEP and TNF were absent from the matrix and could not be tested in the UKB-causal analysis. 359: 360: 6. **Whole-blood non-replication.** The PBMC-derived neuropeptide and GWAS-neural signatures do not replicate in GSE67311 (whole blood), and the mechanism is not explained by neutrophil dilution. The proxies

are PBMC/RNA-seq-specific; plasma protein measurement (Olink/ELISA) is the decisive test. PCSK1N is re-classified as “not confirmed” due to direction inversion. **The absolute expression of the full opioid axis (TACR1/OPRM1/OPRK1/TAC1/PENK) also does not replicate; its co-expression architecture is partially preserved (5 of 10 pairs show FM > HC, rho up to +0.74, but 2 pairs reverse with HC > FM — see §3.3 full 10-pair table)** — suggesting a partially coherent, amplitude-compartment-dependent module. 361: 362: 7. **mRNA ≠ protein.** Elevated mRNA for receptors (TACR1, OPRM1, OPRK1) does not guarantee elevated functional protein. For CA14, the published plasma direction (↓ in CWP) differs from our PBMC mRNA direction (↑), which we interpret as a compartment-specific or compensatory signal; both compartments should be measured. The plasma direction of the neuropeptides remains to be measured (BDNF/NGF precedent: neuropeptides elevated in CSF do not always translate to plasma; review PMC10341963). 363: 364: 8. **Preprint/peer-review status.** Kerrebijn et al. (2025) is published on medRxiv (doi: 10.1101/2025.09.18.25335914, PMID 41001472) and has not yet completed full peer review. The UKB CWP proteomics study is now published (Chen et al., 2025, *Adv Sci*; PMID 41025730; DOI 10.1002/advs.202507691); the peer-reviewed version confirms CA14 causality and adds the direction and agonist information used here. 365: 366: 9. **No experimental validation.** All analyses are computational. No wet-lab or clinical experiments were performed. 367: 368: 9b. **No third PBMC dataset available (2026-08-04).** A systematic GEO search (18 FM series, 4 tissue queries) found only two FM datasets with peripheral blood mRNA transcriptomics: GSE221921 (PBMC RNA-seq, discovery) and GSE67311 (whole blood microarray, non-replicating). A third PBMC dataset for independent replication of the opioid axis does not exist publicly as of 2026-08-04. 369: 370: 10. **Medication, age, and BMI confounding (2026-08-04).** GSE221921 metadata contains only Sample, Etiology, and Gender — no age, BMI, or medication data (the GEO soft file adds only tissue: Blood). This is especially critical for the **opioid axis finding**: FM patients frequently use opioids, antidepressants, and pregabalin, and chronic opioid exposure is known to regulate the expression of opioid receptors themselves. **The elevated OPRM1/OPRK1/TACR1 signal in PBMCs cannot be distinguished from a pharmacological effect with the available data** — it may reflect disease biology, medication, or both. Any validation cohort must record medication status and stratify by it. This limitation is acknowledged for the protocol design in `PROTOCOL_Olink_FM_Biomarker_Validation.md` (§119) and is elevated here to an explicit risk for the current findings. 371: 11. **Tissue Extrapolations of Peripheral Blood mRNA.** Expression levels of *DRD2* measured in peripheral blood cells (PBMCs or whole blood) reflect basal, low-level transcription in circulating immune sub-populations and cannot be interpreted as a direct proxy for central nervous system dopaminergic function, striatal D2 receptor density, or mesolimbic neurotransmission. Central

dopaminergic pathology must be evaluated via brain-specific germline sQTL genetic mechanisms (§4.7) or central neuroimaging/CSF studies. 372: 373: 12. **COL9A1/PTN Bonferroni-survivors characterization corrected (2026-08-04, adversarial review)**. An initial description of “4 robust genes” (COL9A1, BPIFB2, PTN, ST3GAL1) was inflated. Post-adversarial verification: only **COL9A1** ($p_{\text{adj_Bonf}} = 8.5e-5$) and **PTN** ($p_{\text{adj_Bonf}} = 0.020$) survive Bonferroni correction on the sex-adjusted model. BPIFB2 is nominally significant ($p_{\text{adj}} = 0.0035$) but does not survive Bonferroni ($p_{\text{adj_Bonf}} = 0.063$). ST3GAL1 is downregulated ($FC = 0.765$) and anticorrelated with the COL9A1-PTN module ($r = -0.23$). The corrected description: COL9A1 and PTN are the only Bonferroni-surviving sex-adjusted genes from the CWP causal set; BPIFB2 is a secondary candidate; ST3GAL1 is a negative regulator candidate. 374: 375: 12b.

Robustness denominator corrected again (2026-08-04, second pass). Limitation 12 further stated that “no gene passes all 5 models (all fail male-only)” and rescored COL9A1/PTN as “Robust 4/5”. That rescoring silently substituted the model set of `E7_sensitivity_master_GSE221921.csv` — in which a **male-only stratum of 5 FM patients** replaces model 2 — for the five models defined in §2.4. The male-only stratum has no power to reject any null and every gene in the panel fails it, so it cannot function as a robustness criterion. Under the §2.4 definition, which is the one used throughout this manuscript, **COL9A1 and PTN are Robust 5/5**, as are TACR1, OPRM1, TAC1, MDGA2 and DRD2; OPRK1 is 4/5 (female-only $p = 0.056$) and CA14 is 3/5. The two model sets are now explicitly distinguished in §2.4. This correction *raises* the reported robustness of COL9A1/PTN; the substantive constraint on the neuropeptide axis comes from composition adjustment (Limitation 3), not from the model count. 376: 377: — 378: 379: ## 6. Conclusion 380: 381: A targeted reanalysis of two public transcriptomic cohorts, informed by population-scale UK Biobank plasma proteomics, reframes the peripheral molecular signature of fibromyalgia. In FM PBMCs (GSE221921), we find: (1) **an extracellular-matrix / neurite-outgrowth module — COL9A1 ($FC = 2.32$, $d = +0.88$) and PTN ($FC = 2.91$, $d = +0.61$) — is the only signal that survives Bonferroni correction on the sex-adjusted model and adjustment for cell composition**, across all four deconvolution implementations tested; it forms a co-expressed pair ($r = 0.51$) with direct relevance to joint pain (COL9A1, osteoarthritis link) and nerve repair (PTN), is detectable in plasma by Olink, and is **the most robust finding of the investigation** and its primary validation target; (2) a **complete opioid/tachykinin neuropeptide block elevated** (TACR1 $d = +0.60$, OPRM1 $d = +0.53$, TAC1, OPRK1; coordinated co-expression $\rho = 0.31-0.63$) — the largest effect-size block in the investigation, surviving all five sex-sensitivity models but **not** cell-composition adjustment, and therefore best interpreted as a shift in circulating neuropeptide-expressing leukocyte populations rather than per-cell transcriptional activation, with chronic opioid exposure an

unexcluded alternative cause; (3) **CA14, the top UKB causal protein for chronic widespread pain, elevated in PBMC mRNA in the unadjusted analysis** (FC = 2.29, d = +0.41) **but not surviving the sex-stratified primary model** (female-only: p = 0.135, d = +0.24, FC = 1.60; the unadjusted signal is confounded by the structural sex imbalance — see §4.2) **while downregulated in plasma** in the UKB cross-sectional analysis, with MR indicating a protective effect of genetically elevated CA14 — retained here as a direction-specific reference candidate (testable prediction of ↓ CA14 in FM plasma; repurposing direction toward CA14 agonism rather than sulthiame inhibition) rather than the lead target; and (4) **five UKB causal immune-signaling genes downregulated** (TNFRSF1B, CD74, COL18A1, BTN2A1, TNFRSF4), indicating an immunomodulatory/exhaustion pattern rather than classical inflammation — consistent with the absence of IL-6/IL-8 from the highlighted proteins of the largest chronic pain proteomics screen (51,644 UKB participants; TAC1/Substance P not covered by that panel and therefore not evaluable); and (5) the GWAS-prioritized neural genes *MDGA2* and *DRD2* robustly upregulated. These PBMC-derived signatures do not replicate in whole blood (GSE67311), establishing them as PBMC-specific and setting plasma protein measurement as the decisive validation. The dopamine agonist RCT evidence remains limited (one unreplicated positive trial in 21 years). We conclude that the peripheral biology of FM is best described as **extracellular-matrix / neurite-outgrowth dysregulation (COL9A1-PTN) as the cell-intrinsic core, accompanied by a compositional shift in neuropeptide-expressing leukocytes, immunomodulation (UKB immune genes) and pH/nociception dysregulation (CA14)** — and we recommend prioritizing **COL9A1 and PTN** (Olink, plasma-detectable, Bonferroni-surviving, composition-robust) for plasma validation, with CA14 measured as a direction-specific reference, Substance P/enkephalins (ELISA) as secondary neuropeptide measures, and IL-8 retained as an assay control, in sex-balanced, medication-stratified, cell-type-resolved cohorts. Because the neuropeptide signal is compositional, the appropriate follow-up for that axis is cell-resolved (flow cytometry or scRNA-seq) rather than bulk transcriptomic or plasma measurement.

382: 383: — 384: 385: ##
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GWAS-prioritized neural genes vs. mast cell markers across
 PBMC and whole-blood datasets 454: - scripts/
 phase2_rct_review.py — literature review evidence table 455: -
 validate_fm_biomarkers_iter2.py — expanded 19-gene panel with
 Mann-Whitney + Bonferroni + Cohen's d (post-adversarial-audit
 v3) 456: - scripts/validate_fm_biomarkers_gse67311.py — whole-
 blood cross-validation (negative result, verified group counts
 67/75) 457: - scripts/deconvolution_cell_types.py — cell-type
 marker-based deconvolution of GSE67311 458: - scripts/
 validate_opioid_axis_gse67311.py — full opioid/tachykinin axis
 (TACR1/OPRM1/OPRK1/TAC1/PENK) in GSE67311 + co-
 expression Spearman 459: - scripts/
 e1_deconvolution_adjusted_model.py — cell-composition-adjusted
 model (model 6, §2.4): nested M1/M2/M3 fits for the neuropeptide
 axis and COL9A1/PTN, four-implementation variant sweep, and
 the 600-gene negative control 460: - scripts/
 audit_verify_claims.py — executable verifier that recomputes the
 numeric claims of Tables 2 and 4 and the co-expression values
 directly from the raw GEO matrix 461: 462: Representative
 derived tables include: 463: - analysis/
 sensitivity_analysis_GSE221921.csv — five-model robustness table
 for PBMC reanalysis 464: - analysis/
 cross_context_gwas_neural_genes.csv — cross-context comparison
 table used for the PBMC vs. whole-blood contrast 465: - analysis/
 RCT_dopamine_agonists_FM.csv — targeted literature review
 evidence table 466: - analysis/E1b_DECONVOLUCION_COL9A1_PTND.md +
 E1b_variant_sweep.csv + E1b_negative_control_random_genes.csv —
 composition-adjustment results, implementation sweep, and
 negative control 467: - analysis/
 E7_sensitivity_master_GSE221921.csv — alternative model set
 including a male-only stratum; retained for transparency and **not**
 used to score robustness in this manuscript (§2.4) 468: 469:
 Software: Python 3.10, pandas 1.5.3, scipy 1.10.1, statsmodels
 0.13.5. 470: 471: ## AI-Assisted Research Methodology 472: 473:
 This study was conducted using an agentic AI-assisted research
 methodology implemented within a human-in-the-loop (HITL)
 framework. Analytical pipelines, literature synthesis, and
 computational design steps were developed with the support of
 large language model agents (Claude, Anthropic). All scientific
 claims were systematically validated by the human author against
 pre-established, independently verifiable ground truths: AlphaFold
 structural predictions (EMBL-EBI), RCSB PDB crystallographic
 coordinates, ChEMBL bioactivity databases, GEO-deposited
 expression matrices, UK Biobank proteomics publications, and
 published statistical methods. Verification followed an evidence-
 first principle: no finding was accepted without traceable support
 from primary sources. This framework is implemented in the
 open-source agentic-lab-eac package (Apache-2.0, GitHub:
 Grizaceo/agentic-lab-eac), which formalizes the generate-review-
 verify cycle used throughout this study. 474: 475: ## Conflict of
 Interest 476: 477: The author declares no conflicts of interest.
 478: 479: ## Author's Note 480: 481: This study was initiated out

of personal motivation following personal experience with fibromyalgia in close contacts. No financial or institutional interest is involved.